



Universidad Católica
San Pablo

Ciencia de la Computación

Redes y Comunicación

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Definición de Protocolos

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"Los alumnos declaran haber realizado el presente trabajo de acuerdo a las normas de la Universidad Católica San Pablo"

Communication Protocol Specification

Operation: Send Information

Allows the server to send information to clients.

Response: Server → Clients

Field	Size	Description
Type	1 Byte	Packet Type (1: packet)
Number	4 Bytes	The packet sequence number
Size	4 Bytes	Total number of characters in the checksum
Information	Variable	Checksum information for the data to be sent
Number	4 Bytes	Total number of packets
Number	4 Bytes	Packet number
Size	4 Bytes	Size of the data to be sent
Information	Variable	Data to be sent

Allows the client to send information to server.

Request: Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (1: packet)
Number	4 Bytes	The packet sequence number
Size	4 Bytes	Total number of characters in the checksum
Information	Variable	Checksum information for the data to be sent
Number	4 Bytes	Total number of packets
Number	4 Bytes	Packet number
Size	4 Bytes	Size of the data to be sent
Information	Variable	Data to be sent

Operation: ACK

Response: Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (2: ACK)
Number	4 Bytes	The packet sequence number

Operation: NACK

Response: Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (3: NACK)
Number	4 Bytes	The packet sequence number

Usage Examples

Send Information

```
100010256CHECKSUM000500010004HOLA
100020256CHECKSUM000500020005PABLO
100030256CHECKSUM000500030004HOLA
100040256CHECKSUM000500040004Juan
100050256CHECKSUM000500050011Informacion
```

Listing 1: Examples of information sending packets

ACK

```
20001
20002
```

Listing 2: Examples of ACK packets

NACK

```
30001
30002
```

Listing 3: Examples of NACK packets